

Technology Stack

Date	02 July 2026
Team ID	XXXXXX
Project Name	Raising waters
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	<i>HTML5, CSS3, JavaScript, Bootstrap</i>	<i>Used to create a responsive, user-friendly interface where users can enter weather parameters and view flood prediction results.</i>

2	Backend / Server-Side	<i>Python, Flask</i>	<i>Flask provides a lightweight framework to handle user requests, process data, and integrate the machine learning model with the web application.</i>
3	Database / Data Storage	<i>CSV Dataset, Pickle (.pkl) Model File</i>	<i>Historical flood data is stored in CSV format, and the trained XGBoost model is stored as a Pickle file for efficient prediction.</i>
4	Cloud / Hosting / Deployment	<i>IBM Cloud, Flask Local Server</i>	<i>IBM Cloud enables scalable deployment and remote access, while Flask is used for local development and testing.</i>
5	Version Control & CI/CD	<i>Git, GitHub</i>	<i>Git tracks project changes, and GitHub is used for source code management, collaboration, and project submission.</i>
6	Third-Party APIs / Other Tools	<i>Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, Jupyter Notebook, VS Code</i>	<i>These libraries and tools support data preprocessing, model training, visualization, evaluation, and application development. Future versions can integrate a real-time Weather API for live flood prediction.</i>